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command line arguments, fileoperators, read and filters

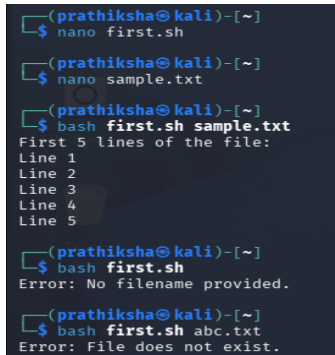
Write a Bash script that accepts a filename as the first command-line argument.

Check whether an argument is provided.

Check whether the file exists using a file operator.

If the file exists, display the first 5 lines using head.

Otherwise, display an appropriate error message.



```
(prathiksha@kali)~  
$ nano first.sh  
(prathiksha@kali)~  
$ nano sample.txt  
(prathiksha@kali)~  
$ bash first.sh sample.txt  
First 5 lines of the file:  
Line 1  
Line 2  
Line 3  
Line 4  
Line 5  
(prathiksha@kali)~  
$ bash first.sh  
Error: No filename provided.  
(prathiksha@kali)~  
$ bash first.sh abc.txt  
Error: File does not exist.
```

```
#!/bin/bash
```

```
if [ -z "$1" ]  
then  
    echo "Error: No filename provided."  
    exit 1  
fi
```

```
if [ -f "$1" ]  
then  
    echo "First 5 lines of the file:"  
    head -n 5 "$1"  
else  
    echo "Error: File does not exist."  
fi
```

File Type and Permission Checker

Write a Bash script that accepts a path as a command-line argument and determines whether it is:

a regular file,

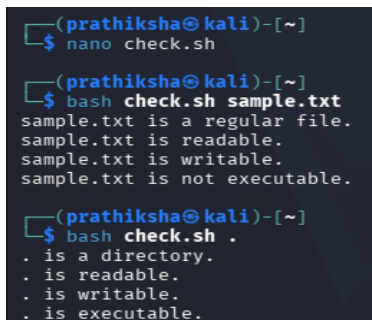
a directory,

readable,

writable, and

executable.

Display suitable messages using file operators such as -f, -d, -r, -w, and -x



```
(prathiksha@kali)~  
$ nano check.sh  
(prathiksha@kali)~  
$ bash check.sh sample.txt  
sample.txt is a regular file.  
sample.txt is readable.  
sample.txt is writable.  
sample.txt is not executable.  
(prathiksha@kali)~  
$ bash check.sh .  
. is a directory.  
. is readable.  
. is writable.  
. is executable.
```

```
#!/bin/bash

if [ -z "$1" ]
then
    echo "Error: No path provided."
    exit 1
fi

if [ -f "$1" ]
then
    echo "$1 is a regular file."
elif [ -d "$1" ]
then
    echo "$1 is a directory."
else
    echo "$1 is neither a regular file nor a directory."
fi

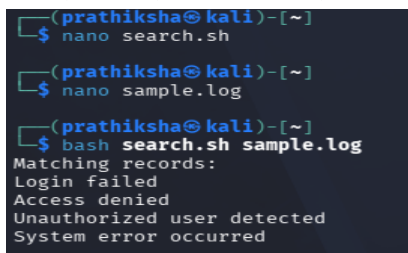
if [ -r "$1" ]
then
    echo "$1 is readable."
else
    echo "$1 is not readable."
fi

if [ -w "$1" ]
then
    echo "$1 is writable."
else
    echo "$1 is not writable."
fi

if [ -x "$1" ]
then
    echo "$1 is executable."
else
    echo "$1 is not executable."
fi
```

-f → regular file
-d → directory
-r → readable
-w → writable
-x → executable

Write a Bash script that accepts a log file as a command-line argument and searches for the words: failed, denied, unauthorized, and error.



```
(prathiksha@kali)~[~]
$ nano search.sh

(prathiksha@kali)~[~]
$ nano sample.log

(prathiksha@kali)~[~]
$ bash search.sh sample.log
Matching records:
Login failed
Access denied
Unauthorized user detected
System error occurred
```

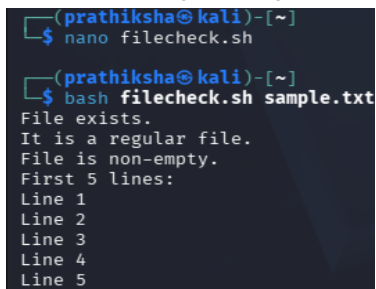
```
#!/bin/bash

if [ -z "$1" ]
then
    echo "Error: No log file provided."
    exit 1
fi

if [ ! -f "$1" ]
then
    echo "Error: Log file does not exist."
    exit 1
fi

echo "Matching records:"
grep -iE "failed|denied|unauthorized|error" "$1"
```

**Write a Bash script that accepts a filename and checks whether:
the file exists,
it is a regular file,
it is empty or non-empty.
If it is non-empty, display the first 5 lines.**



```
(prathiksha@kali)~$ nano filecheck.sh
(prathiksha@kali)~$ bash filecheck.sh sample.txt
File exists.
It is a regular file.
File is non-empty.
First 5 lines:
Line 1
Line 2
Line 3
Line 4
Line 5
```

```
#!/bin/bash

if [ -z "$1" ]
then
    echo "Error: No filename provided."
    exit 1
fi

if [ ! -e "$1" ]
then
    echo "File does not exist."
    exit 1
fi

echo "File exists."

if [ -f "$1" ]
then
    echo "It is a regular file."
else
    echo "It is not a regular file."
    exit 1
fi

if [ -s "$1" ]
then
    echo "File is non-empty."
```

```

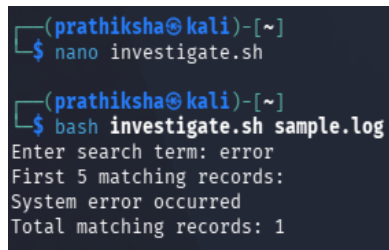
    echo "First 5 lines:"
    head -n 5 "$1"
else
    echo "File is empty."
fi

```

-e → checks whether the file exists
-f → checks whether it is a regular file
-s → checks whether the file is non-empty
head → displays the first 5 lines

Write a Bash script named investigate.sh that accepts a log file as a command-line argument. The script must first verify that the file exists and is readable. Then ask the investigator to enter a search term.

Display the first 5 matching records and report the total number of matching records.



```

(prathiksha@kali)-[~]
$ nano investigate.sh

(prathiksha@kali)-[~]
$ bash investigate.sh sample.log
Enter search term: error
First 5 matching records:
System error occurred
Total matching records: 1

```

```
#!/bin/bash
```

```

if [ -z "$1" ]
then
    echo "Error: No log file provided."
    exit 1
fi

if [ ! -e "$1" ]
then
    echo "Error: Log file does not exist."
    exit 1
fi

if [ ! -r "$1" ]
then
    echo "Error: Log file is not readable."
    exit 1
fi

read -p "Enter search term: " term

echo "First 5 matching records:"
grep -i "$term" "$1" | head -n 5

count=$(grep -ic "$term" "$1")

echo "Total matching records: $count"

```